

Morphogenetic Aspect Networks: Self-Evolving Agents via Reward-Modulated Structural Plasticity over an Externally Reasoning LLM

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Abstract

Aspect-Oriented Programming (AOP) was proposed as a biologically inspired paradigm in which an *aspect* models a neuron, a *pointcut* a content-conditioned routing primitive, and an *advice* a signal transduction; a network of aspects (“aspect-of-aspect”) then models a neural network. When the woven substrate is deterministic code, weaving reliably changes behavior. The contemporary realization of this idea weaves into a *stochastic* large language model (LLM) instead, and the guarantee degrades: most weaving becomes prompt-text injection whose effect the model may or may not honor. We argue that the missing piece is exactly the one the original program left open—*synaptic plasticity*—and that it must be reconceived at the level of *structure*, not weights. We present **Morphogenetic Aspect Networks (MAN)**, a formal nucleus in which an agent is a *stochastic graph-rewriting system*: a typed, dynamic aspect graph whose fast dynamics perform one inference step (routing $\rightarrow$ weaving $\rightarrow$ external-LLM effecting under fixed topology), and whose slow dynamics *grow and reshape the graph itself*. Crucially, both are governed by a single variational functional $F = \text{surprise} + \beta \cdot \text{complexity}$: each structural rewrite fires with rate $\propto \exp(-\Delta F/T)$, so “learning” is gradient-free structure search rather than weight descent on a fixed architecture—a concrete instance of the hypothesis that *topology and learning are unified*. We specify the state, the credit assignment that supplies F ’s error term, and both generative rewrite primitives (*differentiation/split* and *abstraction/lift*) in full, and we prove conservation and behavior-preservation properties that keep self-evolution bounded. Section 9 reports **Phase I internal validity** on graded in-repo fixtures (H1–H5): mechanisms per spec, not application-scale Self-Evolving proof (Phase II deferred).

1 Introduction

From separation of concerns to a nervous system. The separation-of-concerns principle [1] and its realization in Aspect-Oriented Programming [2] modularize *cross-cutting* behavior that object-oriented decomposition cannot localize. An earlier line of work observed that biological nervous systems handle cross-cutting (“non-orthogonal”) concerns more gracefully than mechanistic paradigms do, and proposed AOP as a candidate *biologically inspired* programming paradigm, with the mapping aspect $\leftrightarrow$ neuron, pointcut $\leftrightarrow$ axon, advice $\leftrightarrow$ signal transduction, joinpoint $\leftrightarrow$ synapse, object $\leftrightarrow$ effector, and a three-layer “evolvable software” architecture (organs/spinal-cord/cortex) [3]. That work flagged one consideration as unsolved: *synaptic plasticity*, which it argued would require dynamic AOP.

A note on timing. This mapping was articulated in 2004 [3]—over a decade before content-conditioned routing became central to mainstream architectures through attention [7] and sparsely-gated mixture-of-experts [8], and well before the present wave of LLM agents. The proposal thus anticipated routing-based, conditionally computed architectures; what it lacked was a *substrate powerful enough to serve as the effector*. That effector now exists: a pretrained LLM (together with its action/control surface—tools, memory writes, message sends) is the first general, open-domain competence engine an aspect network can *invoke* rather than hand-program. Crucially, this strength buys *generality, not determinism*: unlike the deterministic effectors of 2004 (a muscle reliably contracts), the LLM is a *strong but stochastic* effector. Reconstructing reliable behavior around such an effector is exactly what MAN exists to do (Section 10)—which is what makes the revival both possible and necessary.

The reliability gap. In the original setting the substrate being woven was deterministic code (static weaving into bytecode), so “weaving reliably changes behavior” held: the joinpoint is a real execution point and the advice is real code. Contemporary agent systems instead weave into an LLM. The effector is no longer a deterministic object whose response to a transmitter is fixed (cf. a neuromuscular junction); it is a stochastic generator. Consequently most advice degrades to text appended to a prompt, whose causal effect is not guaranteed. Empirically, the only *hard* levers are those that intercept the host’s control flow (refuse a tool call, cancel a message, veto a sub-agent); the rest is *soft* influence bounded by the model’s instruction-following.

Thesis. We take the position that the aspect network should be neither another trainable artificial neural network (ANN) nor a second LLM, but a *morphological model of the neuron* in software, committed to two hypotheses: (H-form) *form and function are unified*—the morphology of an aspect is its computational role; and (H-topo) *topology and learning are unified*—unlike an ANN, which separates a fixed differentiable architecture from gradient-based weight search, a morphogenetic substrate learns by *reshaping its own structure*. Under these hypotheses, plasticity is *morphogenesis*: growing, splitting, abstracting, and pruning structure, scored by a single objective. The aspect is well suited to be the atomic unit of a neuro-symbolic system because it is, in one object, both *symbolic* (named, inspectable, composable) and *sub-symbolic* (graded activation, plastic routing).

Contributions.

1. A formal nucleus, **MAN** $M = (N, \Downarrow_{\text{fast}}, \Downarrow_{\text{slow}}, F, \kappa, T(\cdot))$, casting an agent as a stochastic graph-rewriting system in which *a structural rewrite is an inference step in model space* (Section 3).
2. A deterministic, replay-reproducible *credit field* κ combining eligibility traces (temporal), responsibility weights (structural, with a null-player property), and an advantage baseline, obeying credit conservation (Section 4).
3. A single variational selection rule: every rewrite fires with rate $\propto \exp(-\Delta F/T)$ where $F = \text{surprise} + \beta \text{complexity}$, unifying weight and structure updates (Section 5).
4. A full specification of both generative primitives—*differentiation (split)* and *abstraction (lift)*—each with guard, rewrite, and ΔF score, plus conservation and behavior-preservation properties (Sections 6–8).
5. A falsifiable evaluation protocol with preregistered hypotheses, metrics, baselines, and ablations (Section 9).

Honest scope. MAN *orchestrates* an external LLM; reasoning is outsourced to the LLM, while MAN owns structure, governance, and self-evolution. It does not, by itself, form new continuous representations. Its single irreducible weakness (Section 10) is the causal opacity of soft advice over a stochastic effector—the same gap that motivates the framework.

2 Background and Related Work

AOP and process-algebraic semantics. AOP [2] adds aspect, pointcut, joinpoint, and advice to a base paradigm. Its dynamic, mobile communication topology has a natural home in process calculi: CSP [4] and the π -calculus [5], the latter modeling channel names passed as messages (*mobility*)—the formal substrate for a connection graph that changes at runtime. Process-algebraic foundations of AOP were studied by Andrews [6].

Conditional computation and attention. A pointcut performs content-conditioned routing, the same primitive as attention [7] and as the router in sparsely-gated mixture-of-experts [8]. We use this isomorphism at the level of *routing topology*; we do not claim the running system is a trained transformer (Section 10).

Structural learning and morphogenesis. Learning that changes *topology* rather than only weights has a real if minority lineage: neuroevolution of augmenting topologies (NEAT) [9], developmental encodings, and biological accounts of neural morphogenesis and synaptic plasticity [10], with deep roots in morphogenesis itself [11]. These approaches sacrifice the differentiable search space that makes gradient descent powerful—the central tension we inherit (Section 10).

Objectives without gradients. Our selection rule is a free-energy / minimum-description-length tradeoff [12, 13]; structural moves are accepted by a Metropolis criterion. Credit assignment uses eligibility traces and the advantage formulation from reinforcement learning [14, 15], and the Shapley value [16] as the principled (if expensive) responsibility distribution. The fast/slow split mirrors the System-1/System-2 distinction [17] and society-of-mind decompositions [18].

3 The Morphogenetic Aspect Agent

Definition 1 (Agent as a stochastic graph-rewriting system). An agent is a tuple

$$M = (N, \Downarrow_{\text{fast}}, \Downarrow_{\text{slow}}, F, \kappa, T(\cdot)),$$

where N is a typed dynamic aspect graph (state), $\Downarrow_{\text{fast}}$ is one inference step under fixed topology, $\Downarrow_{\text{slow}}$ is the morphogenetic rewrite relation whose moves fire with rate $\min(1, \exp(-\Delta F/T))$, F is a variational functional, κ is the credit field, and $T(\cdot)$ a temperature schedule.

The governing claim is that *structure and inference are one process*: a single application of $\Downarrow_{\text{slow}}$ is a step of inference in model space.

3.1 State $N = (A, S, x)$

- A : aspect nodes. Each a carries a symbolic identity $\tau(a)$, a gain $g(a)$, dendrite/axon ports, and a sliding buffer $D_a = \{(\varphi_t, a_t, r_t)\}$ of context features $\varphi_t \in \mathbb{R}^d$, activation score a_t , and attributed reward r_t .

- S : synapses, directed edges $s = (a_i.\text{axon} \rightarrow a_j.\text{dendrite})$ with weight $w(s) \geq 0$ and eligibility $e(s)$.
- x : the current activation state (“membrane potentials”).

π -calculus reading. An aspect is a process, a synapse a shared channel name, a pointcut a name-matching predicate; mobility (passing channel names) is the formal home of structural plasticity. **Conserved core.** A subset $A_{\text{reflex}} \subseteq A$ of deterministic reflex aspects (inhibitory interneurons) is excluded from random rewriting; it is the invariant boundary (“brainstem”) that keeps self-evolution safe.

3.2 Fast dynamics $\Downarrow_{\text{fast}}$: one inference step

A turn is one π -reduction at fixed topology: (1) joinpoint context φ (the *query*) matches candidate aspects via pointcuts, each receiving an activation score a_i (the attention weight); (2) the weaver composes their advice into an injection under a budget / top- k (the weighted sum of *values*); (3) the external LLM (effector) consumes injection plus context and emits output/actions, while A_{reflex} deterministically gates at the effecting boundary (allow/deny/rewrite). System 1 is the LLM at this step; System 2 is the aspect graph that schedules it.

4 Credit Assignment

A deterministic reward r_t (verifier verdict, effector outcome, task success) is distributed over aspects and synapses by the product of a *temporal*, a *structural*, and an *advantage* factor:

$$\kappa(a) += (r_t - b) e_a(t) \rho_a(t), \quad \kappa(s) += (r_t - b) e_s(t). \quad (1)$$

- **Eligibility e** (temporal): $e_a \leftarrow \lambda e_a + \alpha \cdot \text{part}_a$ (and analogously for edges), handling delayed reward; λ is the credit horizon (synaptic tagging).
- **Responsibility ρ** (structural): tier-1 influence weighting $\rho_i = a_i c_i / \sum_j a_j c_j$ with contribution c_i large for *hard*-acting advice and discounted for *soft* advice; tier-2 sampled counterfactual / Shapley calibration whose null-player property assigns ≈ 0 credit to irrelevant aspects, defeating the “it fired, therefore it gets credit” fallacy.
- **Baseline b** : a context-bucketed running mean, so $r_t - b$ is an *advantage* (a reward-prediction error); credit is proportional to surprise, not raw reward.

Proposition 1 (Credit conservation). With ρ normalized ($\sum_i \rho_i = 1$) and tier-2 satisfying Shapley efficiency, the per-turn node credit is conserved: $\sum_a \kappa_a(t) = r_t - b$. Hence credit is a distribution, not an inflation.

Node credit drives SPLIT/neurogenesis; edge credit drives CONNECT/PRUNE/REWEIGHT. Equation (1) with $\rho \equiv \text{co-activation}$ reduces to reward-modulated Hebbian plasticity with eligibility traces.

5 The Objective F and Slow Dynamics

Definition 2 (Variational objective).

$$F = \underbrace{\text{SURPRISE}}_{\text{task error / verification failure}} + \beta \cdot \underbrace{\text{COMPLEXITY}}_{\text{structural description length}}.$$

The same F scores reweighting, edge creation, and node differentiation alike—there is no separate “architecture search then weight search.” This is the formal content of (H-topo).

Rewrite grammar. Each rule has the form LHS pattern + guard $\Rightarrow$ RHS and fires with rate $\min(1, \exp(-\Delta F/T))$.

Primitive	Effect	Driver
CONNECT (synaptogenesis)	add edge	edge credit $\times$ co-activation
PRUNE	remove edge	low weight, cold
REWEIGHT (LTP/LTD)	change w (degenerate rewrite)	edge credit
SPLIT (differentiation)	one node $\rightarrow$ two specialists	node-credit variance
MERGE (abstraction/fusion)	two near-duplicates $\rightarrow$ one	redundancy
LIFT (aspect-of-aspect)	coalition $\rightarrow$ higher-order aspect	co-firing coalition

SPLIT and LIFT are the *generative core*: the only two that create new symbolic units (differentiation top-down by specialization; abstraction bottom-up by coalition). The rest are housekeeping.

Morphogens. Local sufficient statistics decide *where* to propose moves (high advice variance $\rightarrow$ propose SPLIT; strong coalition co-firing $\rightarrow$ propose LIFT; high co-activation $\times$ credit $\rightarrow$ propose CONNECT), compressing the combinatorial space; ΔF decides *whether* to accept.

The end-to-end loop. Algorithm 1 assembles the state (§3.1), credit (§4), and rewrite grammar above into the self-evolution loop. The agent is *born* as a zygote—the conserved reflex core A_{reflex} plus a single undifferentiated cortex aspect a_0 whose pointcut matches every joinpoint (lines 1–4). Thereafter $\Downarrow_{\text{fast}}$ runs each turn (route, weave, effect, attribute credit), and every P turns $\Downarrow_{\text{slow}}$ proposes morphogen-localized rewrites and accepts each with the Metropolis rate $\min(1, \exp(-\Delta F/T))$. Competence is not edited in by a developer; it is the fixed point that the SPLIT/LIFT dynamics grow—the two generative primitives are specified next (§6, §7).

6 The SPLIT Primitive (worked specification)

SPLIT fires when an aspect is forced to serve two distinguishable sub-contexts demanding conflicting advice. Over the buffer D_a define the reward heterogeneity $H(a) = \text{Var}_t[r_t]$ and, for the best deterministic partition π (axis-aligned threshold stumps over each of the d features plus the leading principal direction; $O(dW \log W)$), the variance-reduction gain

$$G(a) = \text{Var}[r] - (p_1 \text{Var}[r \mid C_1] + p_2 \text{Var}[r \mid C_2]). \quad (2)$$

Guard (all required). $H(a) \geq \theta_H$; $G(a)/H(a) \geq \theta_{\text{sep}}$; $n(a) \geq n_{\text{min}}$; $|\bar{r}_1 - \bar{r}_2| \geq \delta$. The separability ratio rejects splitting on irreducible noise; the sample and effect-size floors reject splitting on chance.

Rewrite (RHS). Replace a by a_1, a_2 with $\text{pc}(a_1) = \text{pc}(a) \wedge \pi$, $\text{pc}(a_2) = \text{pc}(a) \wedge \neg\pi$; route each incoming synapse to the child(ren) whose contexts it drove (split weight when ambiguous, then annealed by REWEIGHT/PRUNE); both children inherit outgoing targets with differentiated gain/advice; children keep named identities $\tau(a) \mid C_1$, $\tau(a) \mid C_2$.

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1: Input: startup directive  $x_0$ ; external LLM effector LLM; reward REWARD; temperature  $T(\cdot)$ ; period  $P$ .
2: Output: evolved aspect graph  $N = (A, S, x)$ .
3: // Genesis: zygote = conserved reflex core + one undifferentiated cortex aspect
4:  $A_{\text{reflex}} \leftarrow \text{SEEDREFLEX}()$  (deterministic, fail-closed; excluded from  $\Downarrow_{\text{slow}}$ )
5:  $a_0 \leftarrow \text{EXTRACT}(x_0)$ ,  $\text{pc}(a_0) \leftarrow \top$  (one intent-aligned cortex aspect; matches every joinpoint)
6:  $N \leftarrow (A_{\text{reflex}} \cup \{a_0\}, \emptyset, x)$  (no synapses at genesis)
7: for each turn  $t$  with joinpoint context  $\varphi_t$  do
8:   //  $\Downarrow_{\text{fast}}$ : one inference step at fixed topology
9:    $A_t \leftarrow \{a \in A : \text{pc}(a) \text{ matches } \varphi_t\}$  with activation scores  $a_i$ 
10:   $\text{inj} \leftarrow \text{WEAVE}(\text{top-}k \text{ of } A_t \text{ by } a_i)$ ;  $y_t \leftarrow \text{LLM}(\varphi_t, \text{inj})$  under  $A_{\text{reflex}}$  gate
11:   $r_t \leftarrow \text{REWARD}(y_t)$ ,  $b \leftarrow \text{BASELINE}(\varphi_t)$  (advantage =  $r_t - b$ )
12:  // credit assignment, Eq. (1) ( $\sum_a \kappa_a = r_t - b$ )
13:  for  $a \in A_t$ :  $e_a \leftarrow \lambda e_a + \alpha \text{part}_a$ ;  $\kappa(a) += (r_t - b) e_a \rho_a$ ;  $D_a \leftarrow D_a \cup \{(\varphi_t, a_t, r_t)\}$ 
14:  for co-active  $s \in S$ :  $e_s \leftarrow \lambda e_s + \alpha$ ;  $\kappa(s) += (r_t - b) e_s$ 
15:  if  $t \bmod P = 0$  then ( $\Downarrow_{\text{slow}}$ : morphogenetic rewrite)
16:     $\text{REWEIGHT}(A, S; \kappa)$ ;  $\text{CONNECT}(S; \kappa)$ ;  $\text{PRUNE}(S)$  (housekeeping; no new symbols)
17:    // generative: morphogen proposes where;  $\Delta F$  decides whether
18:    for  $a \in A \setminus A_{\text{reflex}}$  with  $H(a) \geq \theta_H$  and  $n(a) \geq n_{\min}$  do
19:       $\pi \leftarrow \text{BESTPARTITION}(D_a)$  (axis-aligned, maximize  $G(a)$ ; §6)
20:      if  $\text{SPLITGUARD}(a, \pi) \wedge \text{ACCEPT}(\Delta F_{\text{SPLIT}})$  then  $\text{SPLIT}(a, \pi)$ :  $a \rightarrow a_1, a_2$  with  $\text{pc} \wedge \pi$ ,  $\text{pc} \wedge \neg \pi$ 
21:      for coalition  $K \in \text{COALITIONS}(\text{co-activation graph})$  do
22:        if  $\text{LIFTGUARD}(K) \wedge \text{ACCEPT}(\Delta F_{\text{LIFT}})$  then  $\text{LIFT}(K)$ : coalition  $\rightarrow$  higher-order  $a^*$  (§7)
23:        for near-duplicate  $(a_i, a_j)$ : if  $\text{ACCEPT}(\Delta F_{\text{MERGE}})$  then  $\text{MERGE}(a_i, a_j)$ 
24:        consolidate: lift high-credit aspects into  $A_{\text{reflex}}$ ; archive cold aspects; clear consumed  $D_a$ 
25: return  $N$ 
26: //  $\text{ACCEPT}(\Delta F)$ : true w.p.  $\min(1, \exp(-\Delta F/T))$  (Metropolis;  $T \rightarrow 0 \Rightarrow \text{hard } \Delta F < 0$ )

```

Algorithm 1: Morphogenetic self-evolution: $\Downarrow_{\text{fast}}$ interleaved with $\Downarrow_{\text{slow}}$.

Score. $\Delta \text{COMPLEXITY} = L(\pi) + L(\text{node}) + L(\text{synapses})$ (exact). The benefit term has a cheap log-only estimator $\Delta \text{ERROR} \approx -\eta G(a)$ (variance reduction after credit cleaning, computed by replay), calibrated on the cold path by sampled counterfactual instantiation. Accept iff $\Delta F = \Delta \text{ERROR} + \beta \Delta \text{COMPLEXITY} < 0$.

7 The LIFT Primitive (worked specification)

LIFT is the dual of SPLIT: where SPLIT differentiates one aspect top-down, LIFT abstracts a recurring *coalition* of aspects bottom-up into a higher-order aspect a^* that cross-cuts them—literally an “aspect-of-aspect.” It is the move that creates *depth*.

Candidate coalitions $K = \{a_1, \dots, a_k\}$ are not enumerated over all subsets; they are proposed by cheap community detection / frequent-itemset mining on the co-activation graph (the morphogen of Section 5). For a candidate K define, over the joint window, the *cohesion* $C(K) = P(\text{all } K \text{ active} \mid \text{any } K \text{ active})$ and the *synergy*

$$S(K) = I(\mathbf{1}_K; r) - \sum_i I(a_i; r) \quad (\text{equivalently } R_{\text{joint}}^2 - R_{\text{additive}}^2), \quad (3)$$

the joint reward-information beyond the additive sum of members. $S(K) > 0$ means the coalition carries an *interaction* the parts do not—something worth abstracting.

Guard (all required). $n(K) \geq n_{\min}$ (support); $C(K) \geq \theta_{\text{coh}}$ (they truly co-fire); $S(K) \geq \theta_{\text{syn}}$ (genuine synergy); $2 \leq |K| \leq k_{\max}$; novelty (no existing higher-order aspect already covers K ; otherwise defer to MERGE).

Rewrite (RHS). Create a^* with a symbolic identity $\tau(a^*)$ derived from the coalition members’ names $\tau(a_1), \dots, \tau(a_k)$; its pointcut $\text{pc}(a^*)$ matches when K co-activates (a threshold/conjunction over K). Wire bottom-up aggregation edges $a_i \rightarrow a^*$ and top-down modulation edges $a^* \rightarrow a_i$, seeded from the measured co-activation/synergy. Members are *retained* (unlike SPLIT, which replaces): LIFT adds a layer. **Identity initialization:** a^* ’s coordinating advice/gain starts at the identity (null modulation), so $\Downarrow_{\text{fast}}$ behavior is *unchanged at creation*; a^* acquires influence only later via REWEIGHT gated by ΔF (cf. residual/identity initialization in deep networks).

Score. $\Delta\text{COMPLEXITY} = L(a^*) + L(\text{pc}^*) + L(\text{hub edges}) - L(\text{prunable lateral edges})$: abstraction can *reduce* description length by replacing an $O(k^2)$ entangled clique among members with $O(k)$ hub edges through a^* . The benefit has a cheap log-only estimator $\Delta\text{ERROR} \approx -\eta S(K)$ (the synergy a coordinator can capture), calibrated on the cold path by instantiating a^* with a learned coordinating advice and replaying sampled coalition-active turns. Accept iff $\Delta F < 0$; given identity initialization, the subsequent growth of a^* ’s influence is itself ΔF -gated.

Remark 1 (Lift vs. Merge). Both respond to co-occurrence, but to opposite regimes. High *redundancy* (members duplicate one another; $S(K) \approx 0$, high correlation) triggers MERGE (collapse to one); high *synergy* (members distinct yet jointly load-bearing; $S(K) > 0$) triggers LIFT (abstract a coordinator above them). With SPLIT, the three primitives form a complete response to the three regimes of reward-conditioned co-statistics: separable variance, synergistic coalition, and redundant duplication.

8 Properties

Proposition 2 (Conservation). Every SPLIT satisfies *domain conservation* $\text{dom}(a_1) \uplus \text{dom}(a_2) = \text{dom}(a)$ and *flow conservation* (total in/out weight preserved at creation).

Proposition 3 (Behavior preservation). SPLIT is a *refinement*: since $\text{pc}(a_1) \vee \text{pc}(a_2) \equiv \text{pc}(a)$ and weight is conserved, on any context outside the rewritten region the agent’s $\Downarrow_{\text{fast}}$ behavior is unchanged. LIFT is behavior-preserving *at creation* by identity initialization (its coordinator has null effect until subsequently reweighted under ΔF). Hence both generative primitives change behavior only where doing so lowers F .

Proposition 4 (Replay determinism). The tier-1 quantities (eligibility, responsibility from logged scores and hard/soft contribution flags, baseline, $\Delta\text{COMPLEXITY}$, the variance-reduction estimator) are fixed functions of the activation/reward log. Given the log and constants $(\lambda, \alpha, \beta, T(\cdot))$, the structural trajectory is byte-for-byte reproducible. Tier-2 calibration introduces stochastic LLM replays and is averaged and seeded.

Remark 2 (Three timescales). $\Downarrow_{\text{fast}}$ (per turn) runs inference and accumulates eligibility; a *warm* loop runs REWEIGHT/CONNECT/PRUNE; a *cold* loop runs SPLIT/LIFT/MERGE plus tier-2 calibration and homeostatic normalization. T is annealed (critical periods), and freshly created structure is refractory for τ_{ref} turns to prevent split $\leftrightarrow$ merge thrashing.

9 Evaluation Plan

We preregister five hypotheses (H1–H5) as **Phase I internal validity**: each mechanism works per spec on graded in-repo fixtures. **Phase II** (application-level Self-Evolving in coding/OpenClaw scenarios) is deferred; the auxiliary demo-tool table below is illustrative, not an external-validity claim.

9.1 Hypotheses

- H1 (Efficiency).** As the network matures, the rate of external-LLM calls per solved task decreases (symbolic structure absorbs cognition) without loss of task success.
- H2 (Useful differentiation).** Each accepted SPLIT reduces reward variance within its child sub-contexts relative to the parent, and improves task reward on those sub-contexts.
- H3 (Credit necessity).** Replacing responsibility-weighted credit (Eq. 1) with pure participation increases the spurious-split rate and degrades final reward.
- H4 (Hard>soft reliability).** Behavioral changes mediated by hard (control-flow) aspects are more reliable and more cleanly creditable than those mediated by soft (prompt) aspects; the gap widens with task stochasticity.
- H5 (Bounded self-evolution).** With the conserved core A_{reflex} and refractory/annealing controls, structural size and reward remain stable (no runaway growth or collapse) over long horizons.

9.2 Phase II protocol: self-evolution capability (H0)

H0 (primary, deferred). Through morphogenetic learning alone, with no code changes, the agent’s competence on application scenarios it was *not* hand-built for rises with experience, and the learned structure transfers to held-out and cross-domain scenarios—growing toward general competence rather than overfitting one task. H1–H5 (Phase I) are the mechanism ablations that explain *why* H0 should hold; passing Phase I is necessary but not sufficient for H0.

Learning curve. Success/reward versus experience (episodes) with *no developer edits* mid-run; the signature of self-evolution is a rising curve produced by $\Downarrow_{\text{slow}}$ alone.

Transfer / generalization. Evolve on scenario set A ; evaluate frozen structure on held-out members B and on ≥ 1 cross-domain set. A small $A \rightarrow B$ gap is the testable surrogate for “general”: one substrate and one learning law cover new scenarios *without per-scenario engineering*.

Decisive baseline. Beyond the Phase I baselines, add a *developer-effort-matched hand-iterated* agent (a human iterates prompts/tools under a fixed effort budget). The headline H0 result is MAN’s learned curve overtaking the static graph and approaching the hand-iterated agent at *near-zero developer effort*—the operational form of “no longer iterated traditionally.”

Breadth and cost. Report scenario-family *coverage* (how many distinct families one substrate+law grows into) and *cost-to-competence* (samples/compute to a target success), since sparse-reward credit assignment (Section 10) can make the curve shallow.

Scope. “General” is not claimed literally; it is operationalized as transfer across a diverse, held-out benchmark plus cross-domain generalization, with breadth reported. Target environment: coding / OpenClaw scenarios. **[TODO: fix scenario benchmark, held-out split, cross-domain set, and the hand-iteration effort budget.]**

9.3 Metrics (Phase I)

LLM-calls-per-success; task success / reward; within-child reward variance ratio (pre/post split); *invalid split rate* on noisy bandit (tier-1 splits vs. uniform- ρ no split); reliability gap $\rho_{\text{hard}} - \rho_{\text{soft}}$ and outcome-variance gap under injected noise; structural edge-span on 256-row soak; replay-determinism (re-run from log $\Rightarrow$ identical tier-1 structure). We do *not* treat MERGE counts as a paper primitive (grammar table only).

9.4 Task suites

Graded in-repo suites (regenerate via `scripts/generate_morphogenetic_validation_data.py`): (i) `r_t_bandit.jsonl` (96 rows, partition `zone:alpha/zone:beta`) + `r_t_bandit_noisy.jsonl`; (ii) verifiable tool guard (`demo-tool-block`, 5 commands $\times$ 8 epochs); (iii) `r_t_soak_long.jsonl` (256 rows, $8\times$ bimodal replay). External SWE-bench-scale benchmarks remain future work.

9.5 Baselines and ablations

Baselines: (a) static aspect graph (no $\downarrow_{\text{slow}}$); (b) weight-only plasticity (REWEIGHT alone, ablating SPLIT/LIFT); (c) the underlying LLM with a fixed hand-written prompt; (d) an LLM-only agent with no aspect graph. Ablations: responsibility off (H3); eligibility horizon $\lambda \in \{0, \dots, 1\}$; complexity weight β sweep; temperature schedule on/off; tier-2 calibration on/off; conserved core on/off (H5).

9.6 Phase I results (internal validity, v0.3)

Reproduce: `bash scripts/run-man-paper-experiments.sh`; primary gate file `INTERNAL_VALIDITY.md`; full dump `report.json` (raw `internal_validity`). All H1–H5 + F1–F3 gates pass on current fixtures.

ID	Claim (abbrev.)	Pass
H1	CPS decreases, success held	yes
H2	Split $\downarrow$ variance; bandit lift	yes
H3	ρ spread; tier-1 split on noisy bandit	yes
H4	$\rho_{\text{hard}} > \rho_{\text{soft}}$; noise sweep	yes
H5	Soak span bounded; reflex core retained	yes
F1–F3	Replay, λ , β guards	yes

9.7 Auxiliary tables (demo tool model)

Method	Success $\uparrow$	LLM calls/succ. $\downarrow$	Reliability gap	Struct. stability
LLM-only	0.50	2.00	—	0.00
Fixed hand prompt	1.00	1.00	0.00	0.00
Static aspect graph	1.00	1.00	0.00	0.00
Weight-only plasticity	1.00	1.00	0.00	0.00
MAN (full)	1.00	1.00	0.00	0.00

Ablation	Success / metric	Spurious-split	Notes
– responsibility ρ (H3)	1.00	0.72 vs 0.00	hard–soft spread
– responsibility plasticity	tier-1 split	uniform: no split	noisy + uniform ρ
Tier-1 replay	1.00	—	$ \text{resid} < 10^{-15}$, hash stable
H4 hard>soft (ρ)	gap 0.48	—	tier-1 on tied activations
Tier-2 LOO	splits 0 vs 0	—	Δ split count
– conserved reflex (H5)	stable	edge span 0	reflex on: 2 aspects
H2 bandit lift	1.00	—	$\Delta\bar{r} = 0.5$ post-partition

H1 longitudinal (10 epochs): MAN LLM calls/success $2.0 \rightarrow 1.0$; LLM-only flat at 2.33; static flat at 1.0.

λ	metric	pass
0.00	1.25	yes
0.25	1.67	yes
0.50	2.50	yes
0.75	5.00	yes
0.90	12.07	yes
1.00	40.00	yes

β	metric	pass
0.01	1.00	yes
0.02	1.00	yes
0.05	0.00	no
0.10	0.00	no
0.50	0.00	yes
1.00	0.00	yes

noise	metric	pass
0.00	0.00	no
0.10	0.03	yes
0.20	0.03	yes
0.30	0.05	yes
0.40	-0.02	no

9.8 Threats to validity

Reward attribution quality bounds split quality (Section 10); soft-advice causal estimates are high-variance; feature map φ limits expressible partitions; long-horizon credit is intrinsically hard. The protocol isolates each via the ablations above.

10 Discussion

The central tension. Unifying topology and learning (H-topo) is precisely what removes the differentiable search space that makes gradient descent powerful. MAN therefore searches over

discrete structure, guided by ΔF rather than a gradient. Biology pays for structural learning with evolutionary and developmental time and massive parallelism; the open question is whether morphogen-guided proposals plus a cheap ΔF estimator make it tractable at useful scale.

The irreducible soft spot. The causal effect of a soft advice on a stochastic LLM cannot be known deterministically; counterfactuals only estimate it, with variance. Hence *hard*-acting aspects are cleanly creditable while *soft* ones are statistical. This re-derives, from the credit side, the soft/hard distinction that motivates the framework, and identifies a concrete lever: converting soft advice into hard control-flow mediation improves both reliability and credit cleanliness in one move.

What MAN is and is not. MAN is a stochastic graph-rewriting system whose rewrite rates are set by a variational functional; a readable, governable, growable neuro-symbolic substrate that *orchestrates* an LLM. It is not a gradient-trained ANN/transformer, and it does not aim to out-reason a monolithic model on raw inference; its distinctive value is a transformer one can read, edit, govern, and grow at the architecture level, fusing the organ/spinal-cord/cortex layers of the original program into one continuous substrate.

11 Conclusion

We recast the unsolved “synaptic plasticity” of the aspect-oriented biologically inspired program as *structural* plasticity, and gave a self-contained formal nucleus—Morphogenetic Aspect Networks—in which an agent is a stochastic graph-rewriting system whose structure and inference are one variational process. We specified credit assignment, the variational selection rule, and both generative primitives (SPLIT and LIFT) in full, with conservation and behavior-preservation guarantees, and we preregistered a falsifiable evaluation protocol. The load-bearing open problems are sharp and named: low-variance, low-cost credit estimation, and the acquisition of the feature map φ .

Reproducibility and status

This is a draft framework paper. Sections 3–8 are complete. Section 9 Phase I (internal validity) is filled on current fixtures; Phase II application benchmarks remain future work.

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